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Information handling system keyboard microphone array for audio capture and quality presentation

Abstract

An information handling system keyboard includes an array of microphones having a defined layout that supports location of human voice sounds captured by the array of microphones to isolate voices originating from a defined distance and orientation while filtering other voices and non-voice sounds. A display included in the keyboard depicts a visual indication of voice quality that can include instructions to a speaker regarding adjustments to help obtain a desired voice capture quality to support a video or audio conference of an information handling system interfaced with the keyboard. The keyboard display also supports an interface to control a keyboard, video, mouse (KVM) switch interfaced with the keyboard.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is related to U.S. patent application Ser. No. 17/875,940, filed Jul. 28, 2022,

entitled “Information Handling System Keyboard Display for Keyboard Video Mouse Switch Control,” naming Peng Lip Goh, Suet Chan Law, and Deeder M. Aurongzeb as inventors, which application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates in general to the field of information handling system input devices, and more particularly to an information handling system keyboard microphone array for audio capture and quality presentation.

Description of the Related Art

(3) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

(4) Information handling systems process information with processing components disposed in a housing and interact with an end user through input/output (I/O) devices. Desktop information handling systems generally have a stationary housing that relies upon peripheral I/O devices, such as a peripheral keyboard, mouse and display that communicate through cabled and/or wireless interfaces. Portable information handling systems generally integrate processing components, a display, keyboard and a power source in a portable housing to support mobile operations. Portable information handling systems allow end users to carry a system between meetings, during travel, and between home and office locations so that an end user has access to processing capabilities while mobile. Portable information handling systems typically also support interactions through peripheral I/O devices, which generally provide more intuitive and convenient interactions.

(5) A mouse generally has a housing that fits in an end user's palm and slides on a desktop surface with a position sensor exposed at a bottom surface to detect motion across the desktop surface that is reported to an information handling system as a pointer input and presented at a display as a cursor movement. Two or three buttons are exposed at a top surface to accept “click” inputs. A mouse can experience substantial impact forces over a typical usable life and is therefore generally built in a robust manner having the mouse housing and electronic components assembled with screws. Although assembly with screws provides a durable construction, recycling the mouse generally requires removal of the screws before the plastic of the housing can be crushed and shredded.

(6) Two components of a mouse that tend to wear and fail over time are the scroll wheel and click buttons. A mouse scroll wheel is typically assembled to have a metal spring that contacts against a notched wheel to provide an end user with a tactile feedback of wheel rotation. Over time the spring and notches wear to degrade the tactile feedback. A similar difficulty can arise with the input buttons, which move by a spring or cantilever rotation of a key plate to contact a switch that detects the button input. A cantilever key plate tends to provide a greater button motion to record an input with less precision for the end user to know when the input is made. In contrast, a spring rotation tends to provide a more precise input with less motion of the key plate for high sensitivity with less

of a spring back feel for the end user after the button input is complete. Some of the difficulty in the tactile feel of an input button press is addressed by generation of a click sound made when an input press is performed. In quiet working environments, the click can become a distraction to the end user or others.

(7) Another difficulty that arises with a mouse is that the small size of the housing and the use of the upper surface as the input region makes placement of power and other buttons difficult. Often such buttons are placed in the bottom of the mouse where they are hidden during normal use, however, bottom placement of the button makes its use inconvenient. Further the room available on the bottom surface is minimal and placement of multiple buttons on the bottom can interfere with movement of the mouse on a desktop surface. Generally, a mouse will have a “sleep” mode that reduces power consumption of the mouse when not in use by powering down the processing resource of the mouse and waking when the mouse moves. Although such sleep states decrease power consumption and increase battery life, some power draw is typically involved in monitoring for mouse motion, such as by keeping the position sensor powered up or detection motion with an accelerometer. Further, even in a sleep mode the processing resources tends to dissipate some power.

(8) Often, to help make a workspace more convenient, a mouse and keyboard will interface with a display through a keyboard, video, mouse (KVM) switch. Once an end user configures a keyboard and mouse to interact through the display, the end user can more readily interact in a desktop environment by coupling to the display without having to separately couple or wirelessly interface with the mouse and keyboard. Further, the KVM switch can provide the end user with access to multiple information handling systems through the display so that the end user can switch between systems, such as a desktop and portable system, while using the same peripheral devices with both systems. One difficulty with relying on KVM switches is that the end user generally has to access an onscreen display menu or physical switch of the display to select which information handling system to use.

(9) Other information handling system functions are sometimes coordinated for an information handling system through a variety of different peripheral devices in addition to a keyboard, mouse and display. One example is the use of a camera and microphone to perform communication, such as with a video or audio conference. Often the camera and microphone are integrated in portable information handling systems. These integrated devices are convenient but tend to have lower quality audio and video than peripheral devices. Audio capture in particular can present difficulty since the typical microphone can capture a wide array of sounds near an information handling system. This can result in a low quality conference since general environmental noise and human speaking by non-participants can detract from the listener's ability to discern audio content.

SUMMARY OF THE INVENTION

(10) Therefore, a need has arisen for a system and method which improve mouse recyclability.

(11) A further need exists for a system and method that provides a more durable mouse scroll wheel and input buttons.

(12) A further need exists for a system and method that manages power usage at a mouse.

(13) A further need exists for a system and method that improves audio captured at an information handling system.

(14) A further need exists for a system and method that improves end user accessibility to a KVM switch.

(15) In accordance with the present invention, a system and method are provided which substantially reduce the disadvantages and problems associated with previous methods and systems for manufacture and use of a mouse in association with an information handling system and other peripherals like a keyboard and display.

(16) More specifically, an information handling system mouse is constructed of a plastic chassis having a full perimeter around side that accepts a bottom surface within the perimeter for a clean

appearance and solid construction. A plastic inner frame heat stakes in place for durable construction that survives repeated compressive forces yet is constructed without metal coupling devices, such as screws. A scroll wheel couples into the chassis at the inner frame and includes variable magnetic attraction applied to the scroll wheel as it turns to provide an end user a tactile feedback of scroll wheel rotation. A magnetic focus lens directs the magnetic force towards a ferromagnetic wheel having spikes and valleys so that magnetic force working against scroll wheel rotation varies as the spikes and valley rotate past the magnetic focus lens. Additional adjustments to the variable magnetic force may be provided by adjusting the magnet or magnets that provide magnetic flux to the magnetic focus lens. In one embodiment, electropermanent magnets are included with the mouse key input buttons to provide a user configurable button response, such as a cantilever response having a substantial button movement associated with an input versus a spring response having a small movement. The button switch is located in an acoustic isolation box to selectively adjust audible feedback provided by a switch input.

(17) Mouse power consumption is managed when the mouse is idle with a motion power switch that cuts off power use when the mouse is idle and reestablishes mouse operations by a motion-initiated power application. A conductive ball held within a conductive container has an open circuit that closes when motion of the mouse places the conductive ball into contact with the conductive container so that a processing resource of the mouse can return to an operating state and manage power until the mouse becomes idle again. The power switch is located at a bottom surface of the mouse and includes a channel for vertical travel when placed in the on position so that additional inputs may be made at the power switch through a tactile button input made by vertical travel of the power button from an on position achieved by a sliding motion of the power button.

(18) In addition to the mouse, a keyboard peripheral is provided that enhances audio capture for communication functions and ease-of-use with a KVM switch through control offered by a touchscreen display included in the keyboard. An array of microphones disposed around the perimeter of a touchscreen display included in the keyboard provide a direction and range to spoken voices so that only voices in a defined zone are included in audio communicated through an information handling system conference. The microphones interface with a digital signal processor processing resource that use a first artificial intelligence engine to filter only voices in the defined zone and a second artificial intelligence engine to filter out non-voice sounds. The keyboard display provides real time feedback to the end user of captured sound quality and control over the audio capture parameters. In addition, the keyboard integrated display interfaces with a keyboard, video, mouse (KVM) switch that selects from plural information handling systems which communicates with a keyboard and mouse coupled to the display. The keyboard display provides control of KVM functions normally managed through an onscreen display menu supported by the display scalar. The keyboard display offers a graphical user interface at the end user's fingertips to toggle KVM control between different information handling systems interfaced with the display.

(19) The present invention provides a number of important technical advantages. One example of an important technical advantage is that a mouse having a durable construction is more readily recycled without the use of screws to couple the mouse components together. The mouse scroll wheel provides a tactile feel through magnetic flux that does not suffer from physical wear over time. An end user can select whether to have a cantilever or spring button input feel and also select the amount of audible click feedback provided by a completed input. Power use when the mouse is inactive is eliminated where no power draw is needed to monitor for mouse motion. A display included in a peripheral keyboard manages KVM switch activation and other functions, such as microphone audio for recording end user voice in support of a videoconference. Audio captured by speakers integrated in the keyboard has a high quality and filters out undesired voices and noise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention may be better understood, and its numerous objects, features and advantages made apparent to those skilled in the art by referencing the accompanying drawings. The use of the same reference number throughout the several figures designates a like or similar element.
- (2) FIG. 1 depicts an information handling system that processes information with processing components disposed in a housing and interfaced with a peripheral display, peripheral keyboard and peripheral mouse;
- (3) FIGS. 2, 2A, 2B and 2C, depict a mouse having a unibody chassis constructed without screws;
- (4) FIG. 3 depicts a flow diagram of a process for manufacture and assembly of a mouse without screws or metallic coupling components;
- (5) FIGS. 4, 4A and 4B depict a system for variable scroll wheel spin speed and tactile response;
- (6) FIGS. 5, 5A and 5B depict an alternative system for providing variable scroll wheel spin speed and tactile feel;
- (7) FIGS. 6 and 6A depict an alternative system for providing variable scroll wheel spin speed and tactile feel;
- (8) FIGS. 7, 7A and 7B depict an alternative system for providing variable scroll wheel spin speed and tactile feel;
- (9) FIG. 8 depicts a flow diagram of a process for automatically activating fast and normal scroll wheel rotation;
- (10) FIG. 9 depicts a side cutaway view of an example of a mouse having a push button key plate that configures to cantilever and spring types of behavior;
- (11) FIG. 10 depicts a flow diagram of a process for managing mouse input button cantilever behavior for a push button;
- (12) FIG. 11 depicts a side cutaway view of an example of a mouse having a push button key plate that configures to cantilever and spring types of behavior;
- (13) FIG. 12 depicts a flow diagram of a process for managing mouse input button spring type behavior for a push button;
- (14) FIGS. 13, 13A and 13B depict a power system to wake a mouse from an off power state that avoids passive power dissipation;
- (15) FIGS. 14, 14A, 14B, 14C, and 14D depict a flow diagram and associated circuit diagrams of operation of the motion power switch illustrated in FIG. 13 above;
- (16) FIGS. 15, 15A, 15B, 15C and 15D depict an example of a system configured to manage mouse click sound volume when a mouse input button is pressed;
- (17) FIGS. 16, 16A, 16B, 16C, 16D, 16E, 16F, 16G and 16H depict a mouse sliding button with orthogonal activation;
- (18) FIG. 17 depicts an upper perspective view of a keyboard as an example of plural directional microphones that cooperate to improve an end user audio capture experience;
- (19) FIG. 18 depicts a block diagram of digital signal processing that provides filtered human voice captured in a defined range and field of view of a keyboard having an array of microphones;
- (20) FIG. 19 depicts a flow diagram of a process for managing voice sounds captured at a keyboard microphone array for use in an audio or video conference supported by an information handling system;
- (21) FIG. 20 depicts an example embodiment of presentation at a keyboard integrated display of audio capture conditions for a microphone array integrated in a keyboard;
- (22) FIG. 21 depicts a keyboard having a display that presents low audio capture quality;
- (23) FIG. 22 depicts a flow diagram of a process for indicating at a keyboard display a quality of audio captured by the keyboard microphone array;
- (24) FIG. 23 depicts a block diagram of a system for managing a keyboard, video, mouse (KVM)

switch of a display through a keyboard integrate display;

(25) FIG. **24** depicts an example embodiment of the type of KVM control available through a keyboard touchscreen display;

(26) FIG. **25** depicts another example embodiment of the type of KVM control available through a keyboard touchscreen display;

(27) FIG. **26** depicts another example embodiment of the type of KVM control available through a keyboard touchscreen display;

(28) FIGS. **27A** and **27B** depict a flow diagram of an example embodiment of keyboard touchscreen KVM control and associated use interfaces; and

(29) FIG. **28** depicts an example embodiment of management of dual display and dual information handling system KVM interfaces with a keyboard display.

DETAILED DESCRIPTION

(30) An information handling system mouse and keyboard have enhanced reliability and support a variety of functions, such as audio capture and KVM switch control. For purposes of this disclosure, an information handling system may include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, or other purposes. For example, an information handling system may be a personal computer, a network storage device, or any other suitable device and may vary in size, shape, performance, functionality, and price. The information handling system may include random access memory (RAM), one or more processing resources such as a central processing unit (CPU) or hardware or software control logic, ROM, and/or other types of nonvolatile memory. Additional components of the information handling system may include one or more disk drives, one or more network ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. The information handling system may also include one or more buses operable to transmit communications between the various hardware components.

(31) Referring now to FIG. **1**, an information handling system **10** processes information with processing components disposed in housing **12** and interfaced with a peripheral display **30**, peripheral keyboard **32** and peripheral mouse **34**. In the example embodiment, housing **12** is a desktop housing configured to operate in a fixed location; however, in alternative embodiments a portable housing may be used that integrates a power source and display in the portable housing to operate the system when mobile. A motherboard **14** coupled to housing **12** provides communication between a central processing unit (CPU) **16** that executes instructions to process information and a random access memory (RAM) **18** that stores the instructions and information. An embedded controller (EC) **20** manages operating conditions of the processing components, such as application of power, maintenance of thermal constraints and interactions on a physical level between peripheral devices and CPU **16**. For instance, EC **20** accepts keyboard and mouse inputs made by an end user for communication to CPU **16**. A solid state drive (SSD) **22** provides persistent storage of instructions and information during power down states, such as by storing an operating system and applications in flash memory for retrieval by EC **20** at application of power. A wireless network interface controller (WNIC) **24** provides wireless communication with external networks and peripheral devices, such as through WiFi and BLUETOOTH. A graphics processing unit (GPU) **26** further processes the information to generate visual images for presentation at a display **30**, such as by communicating pixel values through a display cable **28**.

(32) In operation, an end user interacts with the processing components of information handling system **10** through peripheral devices, such as a keyboard **32** that accepts key inputs, a mouse **34** to accept cursor movement and button inputs and a display **30** that presents information as visual images. The peripheral devices may communicate through wireless signals with integrated WNIC's that support BLUETOOTH or other types of wireless personal area networks (WPANs) or may

interface through a cable, such as a USB Type A, B or C port and cable. In addition to presenting visual images, display **30** may accept inputs through a touchscreen, such as a capacitive touch detection surface. As is set forth below in greater detail, display **30** may include a keyboard, video, mouse (KVM) switch that allows keyboard **32** and mouse **34** to interface through display **30** with information handling system **10**. Keyboard **32** has a conventional QWERTY key configuration, although other types of configurations may be used. As is set forth in greater detail below, keyboard **32** may include an LCD that presents a touchscreen interface, such as a number pad or a controller of the KVM switch. Mouse **34** includes a position sensor on a bottom side that detects mouse movements relative to a desktop surface and reports the mouse movements to information handling system **10** to command movement of a cursor presented at display **30**. A pair of buttons on the upper surface of mouse **34** accept end user presses as inputs that are reported to information handling system **10**, such as right click and left click inputs. A scroll wheel located between the buttons rotates forward and backwards to command an up and down scroll of content at display **30**. In various embodiments, various conventional configurations of keyboard **32** and mouse **34** may be used in the present disclosure with the improvements to the art described herein below in greater detail.

(33) Referring now to FIGS. **2**, **2A**, **2B** and **2C**, a mouse **34** is depicted having a unibody chassis **36** constructed without screws. By building mouse **34** without screws, recycling mouse **34** after its useful life is less complex and costly since screws cannot typically be included in a plastic recycle process and must generally instead be manually removed before the plastics are crushed. In the example embodiment of FIG. **2**, mouse **34** is assembled in a chassis with double injection molding of parts that are heat staked or ultrasonic welded together so that screws or any similar metallic coupling devices are not used. In alternative embodiments, other types of plastic couplers may be used, such as adhesives. FIG. **2** depicts an example of mouse **34** having a chassis **36** molded as a single piece of plastic that encompasses a complete perimeter of the body and has a sphere-like palm rest upper surface molded with side walls to create a rigid form factor resistant to high squeeze forces that can result from normal use. A front bridge **42** forms a box-like overall structure that leaves a bottom opening to accept a bottom surface and a central upper surface opening through which left and right main input buttons **46** and a scroll wheel **44** extends. Gaps and steps in chassis **36** are only for moving parts to be exposed, such as left and right main input buttons **46** and side buttons **38**. In the example embodiment, side buttons **38** are integrated with chassis **36** by heat staking or ultrasonic welding while a gap **40** allows some movement of left and right main input buttons **46** relative to chassis **36**. With this arrangement, the only static assembly gap/step is from the bottom surface assembly coupling to chassis **36** at a bottom side where the assembly location is hidden from view for an aesthetically pleasing final appearance. The result of the assembly, as further detailed below, is a high rigidity unibody mouse assembly with minimal assembly gap/steps and ready recyclability.

(34) FIG. **2** depicts an upper perspective view of the fully assembled mouse **34**. FIG. **2A** depicts a bottom perspective view of mouse **34** with the bottom surface **48** coupled to chassis **36** and having a position sensor **50** exposed at a central location to detect motion of mouse **34** relative to a desktop surface. Bottom surface **48** couples within an interior of the perimeter of chassis **36**, which forms a complete circumference around bottom surface **48** for improved structural integrity. The sides of chassis **36** have enhanced thickness in locations that endure the greatest amount of pressure due to squeezing, such as proximate the side buttons. FIG. **2B** depicts a side perspective view of mouse **34** having the input buttons removed and showing a logo **48** and side buttons **38** integrated in chassis **36**. To help ensure recyclability and also used recycled material, logo **48** has ABS and 50% PCR, while chassis **36** and side buttons **38** have ABS plus 65% PCR. FIG. **2C** depicts a bottom perspective view of mouse **34** with the bottom surface removed to expose an inner frame **52** of ABS plus 65% PCR that provides structural support to chassis **36** and for internal components, such as a circuit board that couples the position sensor with a WNIC to communicate mouse inputs

to an information handling system. A light funnel **54** couples to inner frame **52** to direct light from an internal illumination source at logo **48**. Heat stakes **56** at distributed locations of inner frame **52** assemble the inner frame to the chassis without metal coupling devices. Inner frame **52** provides a structure to hold input buttons and the scroll wheel in place. In addition, side buttons **38** couple by heat stakes **56** to chassis **36** to couple the side buttons in place and also add to the structural support of the chassis. Bridge **58** spans across the opening of opposing sides of chassis **36** to complete the perimeter within which the bottom surface of the mouse couples into place.

(35) Referring now to FIG. **3**, a flow diagram depicts a process for manufacture and assembly of mouse **34** without screws of metallic coupling components. The process starts at step **60** with a logo double shot injection molding to integrate a light emissive material in the chassis main material and aligned to accept illumination from the internal light funnel. At step **62**, side buttons are coupled to the chassis with a heat stake that avoids metallic coupling such as screws. At step **64**, the light funnel is heat staked to the chassis aligned to direct illumination at the logo. Once the side buttons and light funnel are coupled in place, the process continues to step **66** to heat state the inner frame to the chassis. The inner frame reinforces the chassis structure strength and to help maintain rigidity without screws or other metallic coupling devices. Once the chassis is complete, the bottom surface is coupled within the bottom chassis opening perimeter and having electrical components assembled in place.

(36) Referring now to FIGS. **4**, **4A** and **4B**, a system for variable scroll wheel spin speed and tactile response is depicted. FIG. **4** depicts a front perspective view of a scroll wheel assembly **68** that rotates to accept scroll inputs from an end user. In the example embodiment, scroll wheel **44** rests in a cradle **72** to rotate about a central axis so that a Hall sensor **70** detects rotation in increments to report to an information handling system as scroll commands, such as by wireless communication. FIG. **4A** depicts a front perspective view of scroll wheel **44** from an opposing side to illustrate a measurement position of Hall sensor **70** relative to a magnet **86** that spins with scroll wheel **44** to provide scroll wheel rotation measurements through change of magnetic flux. In alternative embodiments, alternative types of sensors may be used, such as an optical sensor. Returning to FIG. **4**, scroll wheel **44** couples to a steel wheel **74** having gear-shaped spikes that extend with even spacing about its circumference. Steel wheel **74** spins with rotation of scroll wheel **44** so that the gear-shaped spikes provide variable magnetic force relative to a magnet placed nearby. In the example embodiment, a permanent magnet **78** provides a magnetic flux through a magnetic focus lens **76** that operates against steel wheel **74** to generate a variable spin speed and tactile feedback for rotation of scroll wheel **44**. Magnetic focus lens **76** is, for example a ferromagnetic material that concentrates the magnetic flux proximate spikes of steel wheel **74** so that the impact of the magnetic flux on scroll wheel **44** varies based upon the distance of steel wheel **74** to magnetic focus lens **76** as spikes of the steel wheel pass by the tip of magnetic focus lens **76**. In one example embodiment, magnetic focus lens **76** is an electronic steel having an iron silicon alloy with around 3.2% silicon content to enhance magnetic permeability. A motor **84** rotates a motor wheel **82** engaged with an arm **80** to adjust the amount of magnetic flux applied by the magnetic focus lens **76** to steel wheel **74**.

(37) FIG. **4** depicts permanent magnet **78** in contact with magnetic focus lens **76** to maximize the amount of magnetic flux applied to steel wheel **74** so that a fully available scroll wheel tactile feedback is provided. Motor wheel **82** rotates a pin **88** fit into a slot of arm **80** to a 9 o'clock position to bring permanent magnet **78** in contact with magnetic focus lens **76** for creating the maximal magnetic flux at steel wheel **74**. FIG. **4B** depicts a side view of the scroll wheel system having motor wheel **82** rotated upward to an 11 o'clock position so that arm **80** is pulled by engagement of pin **88** in the slot **90** away from magnetic focus lens **76**, thus creating space between permanent magnet **78** and magnetic focus lens **76**, effectively cutting to zero the amount of magnetic flux applied to steel wheel **74**. In operation, motor **84** turns motor wheel **82** so that pin **88** rotates to a 9 o'clock position to bring permanent magnet **78** in contact with magnetic focus lens **76**.

When an end user spins scroll wheel **44**, steel wheel **74** fixedly coupled to scroll wheel **44** turns synchronously with scroll wheel **44** so that geared-spikes rotate past magnetic focus lens **76**. Spikes that pass near magnetic focus lens **76** generate a greater magnetic attraction than the valley between the spikes due to changes in distance between the magnetic focus lens and the ferromagnetic material of steel wheel **74** based on whether a spike or a valley aligns with magnetic focus lens **76**. Although the example embodiment moves permanent magnet **78** into and out of contact with magnetic focus lens **76** to turn magnetic flux high or low in a binary manner, an alternative embodiment may move magnetic focus lens **76** to have a greater and lesser distance to steel wheel **74** to adjust the magnetic force that operates against rotation of steel wheel **74**.

(38) Referring now to FIGS. **5**, **5A** and **5B**, an alternative system is depicted for providing variable scroll wheel spin speed and tactile feel. Scroll wheel **44** rotates in a cradle **72** with a steel wheel **74** having geared-spikes coupled in a fixed manner as described above. Steel wheel **74** may be made of any high magnetic permeable material, such as various ferromagnetic alloys including silicon iron alloys, and may have other types of forms that vary the distance of the steel wheel material to a magnetic lens during rotation with the minimum to maximum distance occurring in a manner associated with a desired scroll wheel tactile feel or response. A focus lens assembly **92** has upper and lower arms extending towards steel wheel **74** and a permanent magnet **98** coupled between the upper and lower arms. FIG. **5A** depicts the magnetic field generated between the upper and lower arms of focus lens assembly **92**, which operates with a variable magnetic attraction as steel wheel **74** rotates to align spikes and valleys with the upper and lower arms. Focus lens assembly **92** has a spacing between the upper and lower arms so that spikes and valleys align synchronously with the upper and lower arms as steel wheel **74** rotates. When both the upper and lower arms of focus lens assembly **92** align with spikes of steel wheel **74**, magnetic attraction is greatest; and when both the upper and lower arms of focus lens assembly **92** align with a valley between the spikes of steel wheel **74**, magnetic attraction is least. A motor **96** couples to a rotating arm **94** that supports a second permanent magnet **100**. FIG. **5A** depicts second permanent magnet **100** displaced from alignment with focus lens assembly **92** so that the magnetic field **102** of first magnet **98** operates through the extending arms of focus lens assembly **92** on steel wheel **74**. FIG. **5B** depicts second permanent magnet **100** rotated by rotating arm **94** and motor **96** to align within focus lens assembly **92**. In the example embodiment, first and second permanent magnets **98** and **100** have opposite polar alignments as depicted by FIG. **5B** so that alignment of second permanent magnet **100** within focus lens assembly **92** has the effect of turning off magnetic field **102** attraction operating against steel wheel **74**.

(39) Referring now to FIGS. **6** and **6A**, an alternative system is depicted for providing variable scroll wheel spin speed and tactile feel. In the example embodiment, scroll wheel **44** couples with steel wheel **74** having geared-spikes that vary magnetic attraction relative to a permanent magnet **108** having a Halbach array. As is described above, the distance between spikes and valleys of the steel wheel **74** and permanent magnet **108** varies as scroll wheel **44** rotates to provide the variable spin speed and tactile feel. The Halbach array provides a magnetic attraction with north and south poles on the side of the magnet facing steel wheel **74** and minimal magnetic flux on the opposing side. In one example embodiment, the Halbach array magnetic poles are spaced so that each pole simultaneously aligns with a spike or a valley, such as with an interval of the spikes of steel wheel **74**, so that magnetic attraction is maximized at alignment of the spikes and minimized at alignment of the valleys. A swinging arm **106** couples through a rotating disk **104** with a motor **84** that changes the distance between permanent magnet **108** and steel wheel **74**. FIG. **6** depicts motor **84** engaging swinging arm **106** with rotating disk **104** to place permanent magnet **108** in close proximity to steel wheel **74** so that magnetic attraction is maximized for generating scroll wheel **44** tactile feedback. FIG. **6B** depicts motor **84** rotating swinging arm **106** through rotating disk **104** to move permanent magnet **108** away from steel wheel **74** so that scroll wheel **44** has reduced magnetic interaction for a free spinning operation.

(40) Referring now to FIGS. 7, 7A and 7B, an alternative system is depicted for providing variable scroll wheel spin speed and tactile feel. In the example embodiment, steel wheel **74** couples to scroll wheel **44** as described above so that spikes and valleys of steel wheel **74** interact with magnetic force of a magnetic focus lens assembly **110** to vary scroll wheel rotation speed. As with the embodiment described above with respect to FIG. 5, magnetic focus lens assembly **110** is built with magnetic permeable material, such as ferromagnetic material like a silicon iron alloy, and has an upper and lower arm that each extend towards steel wheel **74** to alternatively align with spikes and valleys that adjust the magnetic force to vary scroll wheel speed. In the example embodiment, a shaft **112** couples to a geared wheel **114** at a base and to a pair of support arms **118** that each hold a permanent magnet. A first permanent magnet couples in place within magnetic focus lens assembly **110** having a first polarity, such as north at top and south at bottom. Geared wheel **114** engages with a rotating wheel assembly **116** and motor **84** so that operation of motor **84** rotates shaft **112** to select which of the pair of support arms **118** brings a magnet into alignment with magnetic focus lens assembly **110**. FIG. 7A depicts alignment of a second permanent magnet **118** having the same polarity orientation as the first permanent magnet with magnetic focus lens assembly **110** so that the combined magnetic attraction of both the first and second magnets operate against the steel wheel **74**, as shown. FIG. 7B depicts wheel **114** rotated to rotate shaft **112** so that a third magnet **120** with an opposite polarity orientation of the first magnet aligns with magnetic focus lens assembly **110** instead of the second magnet **118**. The opposite polarity configurations of the first and third magnets aligned with magnetic focus lens assembly **110** cancels the magnetic attraction directed towards steel wheel **74** so that scroll wheel **44** rotates freely.

(41) Referring now to FIG. 8, a flow diagram depicts a process for automatically activating fast and normal scroll wheel rotation. The process starts at step **122** with power applied to the mouse and at step **124** the mouse detects scroll wheel rotation provided with normal tactile feedback by magnetic attraction proximate the steel wheel that changes as spikes and valleys pass by the magnet as the steel wheel rotates synchronously with the scroll wheel. At step **126**, the Hall sensor or other position sensor detects scroll wheel rotation, such as by sensing movement of a magnet coupled to the scroll wheel and synchronously rotating with the scroll wheel. At step **128**, the Hall sensor or other position sensor detects a pulse rate associated with the rotation of the scroll wheel and indicative of the scroll wheel rotation rate. At step **130**, a processing resource of the mouse, such as a microcontroller unit (MCU) receives the pulse rate to report the scroll wheel input to an information handling system. At step **132**, a determination is made of whether the pulse rate exceeds a threshold pulse rate associated with a fast scroll rate. If not, the process returns to step **128** to continue monitoring scroll wheel rotation.

(42) If at step **132** the fast scroll threshold is exceeded, the process continues to step **134** at which the processing resource commands the motor to rotate from the normal scroll tactile feedback position to a fast scroll position. At step **136**, rotation by the motor moves the permanent magnet relative to the magnetic focus lens to remove the magnetic field from interaction with the steel wheel. As set forth above, the movement may pull the magnet out of alignment with the magnetic focus lens or may introduce an opposing polarity magnet to cancel the magnetic attraction. At step **138**, the mouse scroll wheel will spin freely without the tactile feedback in the absence of the magnetic field. At step **140** a determination is made of whether the pulse rate detected by the scroll wheel rotation has dropped below the fast scroll threshold for a predetermined time. If not, the process returns to step **138** to continue monitoring the pulse rate. If the pulse rate has dropped below the threshold, the process continues to step **142** at which the processing resource detects the lower pulse rate. At step **144**, the processing resource commands the motor to rotate the magnet back into proximity with the magnetic focus lens. At step **146**, the permanent magnet returns to proximity with the magnetic focus lens to provide magnetic attraction that works against the steel wheel and the process returns to step **124**.

(43) Referring now to FIG. 9, a side cutaway view depicts an example of a mouse **34** having a push

button **46** key plate that configures to cantilever and spring types of behavior. In the example configuration of FIG. **9**, a cantilever type of push button behavior is provided with a relatively large rotational movement about a pivot couple **150** to have a keyplate protrusion **152** reach a switch plunger **154** of a switch **156** and press downward to perform an input. To provide an upward bias to button **46** a permanent magnet **162** coupled to button **46** has an opposite pole configuration indicated by arrow **164** relative to an electropermanent magnet **160** coupled to a mouse main board having a power source **166**. The opposite pole configuration provides a spacing between permanent magnet **162** and electropermanent magnet **160** that a press of push button **46** must overcome to engage key plate protrusion **152** against switch plunger **154**. The amount of upward bias may be adjusted by increasing or decreasing the opposing magnetic field of electropermanent magnet **160**. Generally, an electropermanent magnet is built with a permanent magnet having a pole configuration and a secondary magnet having a coil **158** that accepts current to change the polarity of the secondary magnet. When the permanent magnet and secondary magnet have the same polarity, the magnetic field is cumulative of both magnets. When the permanent magnet and secondary magnets have opposite polarity, the magnetic fields cancel each other so that the external magnetic field is near zero. In the example embodiment, the magnetic attraction or repulsion provided by the electropermanent magnet **160** is managed by a processing resource that sets current at one or more coils to adjust one or more magnetic fields in an array of permanent magnets and electromagnets, such as with one or more MCU GPIO pins. In alternative embodiments, other adjustments to the bias working upwards at button **46** may be used, such as electromagnets or mechanical springs.

(44) Referring now to FIG. **10**, a flow diagram depicts a process for managing mouse input button cantilever behavior for a push button. The process starts at step **168** with power up of the mouse. At step **170** a pulse of current is applied to the electropermanent magnet coil to adjust the magnetic pole configuration and turn “on” the electropermanent magnet. At step **172**, the current applied to the coil adjusts the magnetic pole configuration of the magnet array of the electropermanent magnet so that both magnets of the electropermanent magnet array have the same polarity, which is also the same polarity of the permanent magnet coupled to the mouse push button and opposed to the electropermanent magnet. At step **174**, the permanent magnet of the push button and the electropermanent magnet oppose each other to create a repelling force that pushes the key plate of the push button upward and away from the input switch. At step **176**, a determination is made of whether the key plate repulsing force provides sufficient spring back force for the desired cantilever behavior of the input button. If yes, the process ends at step **180**. If not, the process continues to step **178** to increase the current on the electropermanent magnet so that a stronger repulsing force is created and then returns to step **174**. Setting the repulsing force may be performed manually by an end user to a desired preference response, or automatically based upon how presses are performed at the mouse button.

(45) Referring now to FIG. **11**, a side cutaway view depicts an example of a mouse **34** having a push button **46** key plate that configures to cantilever and spring types of behavior. In the example configuration of FIG. **11**, a spring type of push button behavior is provided with a relatively small rotational movement about a pivot couple **150** to have a keyplate protrusion **152** reach a switch plunger **154** of a switch **156** and press downward to perform an input. In the example embodiment, magnetic attraction is established by configuration of the magnetic polarity of electropermanent magnet **160** with current from a power source **166** to a coil **158** so that an attractive magnetic force is provided relative to permanent magnet **162**. The amount of magnetic attraction is sufficient to bring key plate protrusion **152** downward against switch plunger **154** of switch **156** without actuating a switch input by pressing down on switch plunger **154** with sufficient input force. The attractive magnetic force reduces the gap between key plate protrusion **152** and switch plunger **154** to zero so that an input force by an end user pressing at push button **46** will result in an input click with minimal push button movement. Although the example embodiment depicts opposite

magnetic poles to bring member **152** towards switch **156**, a neutral ferromagnetic material may be sufficient for the magnet **162** to attain the desired position without activating the opposite pole of magnet **160**.

(46) Referring now to FIG. **12**, a flow diagram depicts a process for managing mouse input button spring type behavior for a push button. The process starts at step **182** with power up of the mouse. At step **184** a pulse of current is applied to the electropermanent magnet coil to adjust the magnetic pole configuration and turn “on” the electropermanent magnet with a reverse current and polarity direction from that of FIG. **10**. At step **186**, the current applied to the coil adjusts the magnetic pole configuration of the magnet array of the electropermanent magnet so that both magnets of the electropermanent magnet array have the same polarity, which is also the opposite polarity of the permanent magnet coupled to the mouse push button and opposed to the electropermanent magnet. At step **188**, the permanent magnet of the push button and the electropermanent magnet oppose each other to create an attracting force that pulls the key plate of the push button downward and towards the input switch. At step **190**, a determination is made of whether the key plate attracting force provides sufficient or excessive force for the desired spring type behavior of the input button. For instance, the attractive force should keep the amount of movement to make an input at the push button minimal but not so excessive that inadvertent inputs will result. If yes, the process ends at step **194**. If not, the process continues to step **192** to increase or decrease the current on the electropermanent magnet so that a stronger or weaker attracting force is created as desired and then returns to step **188**. Setting the attracting force may be performed manually by an end user to a desired preference response, or automatically based upon how presses are performed at the mouse button.

(47) Referring now to FIGS. **13**, **13A** and **13B**, a power system is depicted to wake a mouse from an off power state that avoids passive power dissipation. FIG. **13** depicts the motion power switch **202** coupled to a mouse printed circuit board **200** within a transparent view of mouse **34**. Generally, mouse **34** has a processing resource on printed circuit board **200**, such as an MCU, that commands a sleep state when mouse **34** meets predetermined conditions of nonuse, such as resting without motion for a predetermined time. In one sleep state, movement detected by a position sensor wakes the mouse, such as with a GPIO signal to the processing resource. As an alternative another sleep state is monitored by an accelerometer that wakes the processing resource when movement is detected. As yet another alternative, another sleep state puts all resources to sleep and wakes when a physical input is made by the end user, such as a button press or scroll wheel rotation. Each of these sleep states use some level of power, either due to power applied to the sensor that detects motion or some minor level of power consumed by the processing resource in a low power sleep state due to passive power dissipation. Motion power switch eliminates all power consumption in a sleep state and powers up mouse **34** when motion is detected to allow an end user to continue mouse use similar to the use case of a conventional sleep state. When the predetermined sleep conditions are detected, the processing resource commands the power supply to switch off, eliminating all power dissipation and preventing return to a power on state by a command of the processing resource. When motion is detected by motion power switch **202**, power is applied to the processing resource so that the processing resource can determine whether to continue sleeping or return to the power on state.

(48) FIG. **13A** depicts a side perspective view of motion power switch **202** mounted on printed circuit board **200**. A nonconductive holder **208**, such as plastic material, couples to printed circuit board **200** and contains a conductive cap **204** that encloses a conductive ball bearing **206**. FIG. **13B** depicts a side cutaway view of motion power switch **202** having a central conductive cup **210** on which conductive ball bearing **206** rests. Conductive cap **204** has a conductive wireline or structure **212** that passes through printed circuit board **200** so that both conductive cup **210** and conductive cap **204** form an incomplete or open circuit that conductive ball bearing **206** can close when motion of mouse **34** works to press conductive ball bearing **206** against both conductive cup **210** and

conductive cap **204**. When the circuit of motion power switch **202** closes, power is applied to the processing resource so that the mouse is powered for use and the processing resource can monitor the use to determine if the power up or power down state is appropriate. In the example embodiment, the conductive cup, ball bearing and cap are a steel and/or copper material, although other materials may be used as appropriate. In the off state of the mouse, current is made available at conductive cup **210** to flow through conductive ball bearing **26** to conductive cap **204** when movement causes the conductive ball to contact both the conductive cup and the conductive cap; applying current at conductive cup **210** limits the risk of an inadvertent ground to the conductive cap causing a complete circuit, however, current may be applied in an opposite direction. Only a brief completion of the circuit is needed to switch power back on to the processing resource so that the processing resource can determine whether to continue applying power or switch back off.

(49) Referring now to FIGS. **14**, **14A**, **14B**, **14C**, and **14D**, a flow diagram and associated circuit diagrams depict operation of the motion power switch **202** illustrated in FIG. **13** above. Generally, a power connection to the mouse processing resource **232**, labeled as an exemplary MCU in the figures, at mouse position tracking sensor **234** has two paths. One path is a normal operating path enabled by end user selection of power on with a power switch **238**. The other path is through the motion power switch enabled when motion closes the circuit by touching the ball bearing against the conductive cap. When the mouse is stationary, control of the power connection switches to the motion power switch **242**, which has an open circuit, while the rest of the mouse powers off to eliminate power consumption. When the mouse moves, the motion power switch closes the power circuit to power up the entire mouse, including the MCU processing resource, momentarily. Once awake, the MCU processing resource switches back to the normal operating path until another power down is appropriate due to lack of motion. Although not depicted in the example, logic of the processing resource monitors the motion power switch for a stuck on position, such as might happen at a tipped desktop surface, to apply conventional sleep logic where the motion power switch motion indication is not consistent with detected mouse use.

(50) At step **220** of FIG. **14**, the process detects movement of the mouse, such as with the position sensor, an accelerometer and the motion power switch. With reference to FIG. **14A**, in response to detection of mouse movement, switch A labeled **236** closes to directly power the MCU processing resource **232** and position sensor **234**. Switch B labeled **240** is commanded open to remove power detection signals of motion power switch **242**, which may periodically close in response to movement but has no power applied. At step **222** a determination is made of whether movement is detected at the mouse, such as by monitoring positions reported by position sensor **234**. If motion is detected, such as with logic of the MCU processing resource, the process returns to step **220** to continue normal use of the mouse in the power on state. If motion is not detected at step **222** for a predetermined time, the process continues to step **224** to enter a low power state. With reference to FIG. **14B**, at step **224** switch A labeled **236** is commanded to an open position so that power is removed from the processing resource and position sensor and switch B labeled **240** is closed to enable monitoring by the motion power switch **202**. With no motion applied to the mouse, the lack of power provided through motion power switch **202** powers down the entire mouse.

(51) At step **226**, the mouse is monitored in the power off state with the motion power switch so that no power is consumed until motion is detected. Once motion is detected, the process continues to step **228** to return operative control of the mouse to the MCU processing resource **232**. With reference to FIG. **14C**, the ball bearing movement within the motion power switch **242** closes the circuit to provide power to position sensor **234** and MCU processing resource **232**. With completion of the power circuit, MCU processing resource **232** confirms movement of the mouse as detected by position sensor **234**. At step **230** and with reference to FIG. **14D**, switch A labeled **236** is closed to provide normal operating power conditions at the mouse so that MCU processing resource **232** can monitor use and control the power state. Switch B labeled **240** is opened to remove power control from motion power switch **242**. Operational control of the mouse remains

with MCU processing resource **232** and the process returns to step **222** to continue monitoring whether motion is detected at the mouse.

(52) Referring now to FIGS. **15**, **15A**, **15B**, **15C** and **15D**, an example is depicted of a system configured to manage mouse click sound volume when a mouse input button is pressed. FIG. **15** depicts mouse **34** with a transparent perspective view that illustrates an example of an acoustic isolation system **250** to manage the audible feedback of mouse input clicks translated from an upper surface input button through a member **262** to press an input switch within the mouse housing. Acoustic isolation system **250** has a stationary front half coupled to a sliding rear half that adjusts the space for sound to exit from the switch. An insulating layer couples to the inner wall of each half to absorb sound and block noise from propagating out the side of the material while an opening above the switch, when space is created by sliding the rear half from the front half, provides a sound chamber exit that can amplify and direct the sound out when desired. An end user selects the amount of audible feedback with an external selector that moves the rear half of the system to close or open the space above the switch.

(53) FIG. **15A** depicts a detailed perspective view of acoustic isolation system **250** having a rear half **252** slid to a forward position as indicated by arrow **256** relative to a front half **254**, which has a fixed position relative to a switch **264** held within the enclosed space. With the rear half **252** slid full forward, a small opening **258** is provided through which member **262** extends down into the interior to engage with switch **264**. The opening **258** may have insulation and a rubber or other seal that aids in attenuation of sound created by switch actuation from leaving the acoustically isolated area. FIG. **15B** depicts a detailed perspective view of acoustic isolation system **250** having a rear half slid rearward to provide an opening **260** above a contained area within rear half **252** that amplifies the sound generated by switch actuation in an echo chamber and directs the sound out from the mouse. FIG. **15C** depicts a side cutaway view of acoustic isolation system **250** in an open position so that switch sounds are directed out from the mouse. Member **262** enters through the front opening to engage against an input plunger **268** of switch **264** so that presses of an upper button surface on the mouse chassis translates to a mouse button “click” input. The area behind switch **264** echoes the click sound and, with the rear half slid rearward to open the area behind the switch, directs the sound up and out from the mouse. An externally exposed sliding handle **266** couples to the rear side of acoustic isolation system **250** so that an end user can select a quiet operating mode or an audio feedback operating mode. FIG. **15D** depicts sliding handle **266** slid to a forward position so that the rear side **252** slides forward to close the rear opening and locate proximate member **262** to leave minimal space for escape of audible sounds.

(54) Referring now to FIGS. **16**, **16A**, **16B**, **16C**, **16D**, **16E**, **16F**, **16G** and **16H**, a mouse sliding button with orthogonal activation is depicted. FIG. **16** depicts a bottom perspective view of mouse **34** with a transparent view of bottom surface **48** showing the interaction of a dual-action power button **274** with a printed circuit board **276**. The dual-action power button **274** slides within a cavity of bottom surface **48** between an off position and an on position to turn power on and off at mouse **34**. In addition, once dual action power button **274** slides to an on position, a press input becomes available as an additional input that can activate functions at mouse **34**, such as wireless advertising or changes to mouse position sensor sensitivity. FIG. **16A** depicts a detailed perspective view of the sliding power button **278** that translates sliding movement as an input to a sliding switch **280**. FIG. **16B** depicts a transparent detailed perspective view of sliding power button **278** to illustrate the location of a tactile switch **282** under sliding power button **278** that is accessible through vertical movement of sliding power button **278** once it has slid to the on position. While mouse **34** remains on with the sliding power switch in the on position, a press inward at the sliding power button provides an input at tactile switch **282**. Once the sliding power switch slides to off, the tactile switch **282** is prevented from actuation. The combined placement of both switches in one location saves space at the bottom surface **48** for more efficient interactions. Although the example embodiment uses a tactile switch for actuation after sliding the power switch on, alternative

embodiments may use alternative switch interfaces that activate based upon a sliding power switch slid to an on position. As an example, sliding the power switch on may expose a capacitive sensor to detect a secondary input that is confirmed with haptic feedback. Other types of buttons may be used.

(55) FIG. 16C depicts a bottom view of the dual position power button in an off position and having a cross-sectional view indicated as shown by FIG. 16D. In the off position, sliding power button 278 is captured under a button holder 286 lip to prevent an inward motion of sliding power button 278 while allowing a sliding movement. Sliding switch 280 interfaces at an actuator 290 with sliding power button 278 so that sliding movement provides an input to power up and power down the mouse at power switch 280. Tactile switch 282 couples to a printed circuit board 276 within a cavity formed by sliding power button 278, as is illustrated by the cross-sectional view of FIG. 16D. A spring plate 284 is configured to bias downward against sliding power button 278 to prevent an inadvertent input to tactile switch 282, although in the off position shown by FIGS. 16C and 16D, the lip of button holder 286 prevents an inward vertical movement of sliding power switch 278 that would actuate tactile switch 282.

(56) FIG. 16E depicts a bottom view of the dual position power button in an on position and having no actuation of tactile switch 282, and a cross-sectional view indicated by FIG. 16F. To slide to the on position, sliding power button 278 moves as indicated by arrow 292 within a cavity of bottom surface 48. In the on position, the rear edge of sliding power button 278 is released from under button holder 286 lip, yet biased downward by spring 284 so that tactile switch 282 is not actuated. Sliding motion of sliding power switch 278 moves power switch 280 actuator 290 to the on position and aligns a ramp 288 formed in sliding power switch 278 with an open area above so that sliding power switch 278 is free to move inward if pressed upward by an end user from the mouse bottom surface. FIG. 16G depicts a bottom view of the dual position power button in an on position and having a press inward to actuate tactile switch 282. The upward press causes a lift up of sliding power button 278 at the rear side as indicated by arrow 294 so that a rotational movement as indicated by arrow 296 results in orthogonal triggering of tactile switch 282. Cross-sectional view depicted by FIG. 16H depicts an upward and inward push of sliding power button 278 as indicated by arrow 294 to overcome the downward bias of spring 284 so that tactile switch 282 is actuated. A range of pivot rotation as indicated by arrow 296 is set based upon engagement of ramp 288 against the printed circuit board 276 bottom surface. Spring 284 returns sliding power button 278 to a neutral position shown in FIG. 16F when the input is released by the end user.

(57) Referring now to FIG. 17, an upper perspective view of a keyboard depicts an example of plural directional microphones that cooperate to improve an end user audio capture experience. In the example embodiment, keyboard 32 has three microphones 300, 302 and 304, configured to capture end user speech along a defined reception range and field of view while dropping sounds that originate outside the defined reception range and field of view. As an example, microphones 300, 302 and 304 cooperate through a processing resource to limit captured sounds to those within 120 degrees of the front side of keyboard 32 and having a range of one meter or less. Sounds of people talking off of the defined field of view, such as the sounds indicated by lines 306, or outside of the defined one meter range, are dropped so that an end user engaged in a videoconference will have only desired audio communicated over the videoconference. A display 346 coupled in keyboard 32 provides a visual indication to an end user of the captured audio, as described below, and allows the end user to adjust the microphone array audio capture, such as by changing the defined range and field of view in which audio is captured. For example, display 346 includes a touchscreen to directly launch a microphone array application, track microphone array performance and adjust audio capture definitions.

(58) In the example embodiment, microphones 300, 302 and 304 are placed in an inverted triangle configuration around the periphery of display 346 at least 50 mm horizontally or diagonally spaced. A processing resource, such as a digital signal processor (DSP), interfaces with the microphones to

monitor captured sounds at each microphone using triangulation to identify sounds that originate in the defined range and field of view. When an end user speaks in front of keyboard **32**, each microphone captures the audio of the end user voice with a difference in time of capture for each microphone based upon the location of the end user to each microphone. As an example, the DSP bases the difference in range of each microphone to the source of the speech based upon an analysis of the wave phases detected for each microphone, such as with an artificial intelligence algorithm. The distance for each microphone provides triangulation to the source of captured audio for comparison with the defined capture distance and field of view. In one embodiment, a second DSP receives the captured audio identified as being in range and field of view or out of range and field of view, and then filters the desired audio to remove audio associated with capture outside of the defined range and field of view. The filter may include separate filter steps to remove out-of-range human voice and to remove non-human sounds completely.

(59) Referring now to FIG. **18**, a block diagram depicts digital signal processing that provides filtered human voice captured in a defined range and field of view of a keyboard having an array of microphones. In the example embodiment, a microphone array **310**, such as the three microphone array shown in FIG. **17**, detects audible sounds and provides the audible sounds to a first processing resource **312** within an audio processing system **308**. For example, processing resource **312** is a first digital signal processor (DSP) having artificial intelligence human voice analysis that calculates and determines voice angle and distances for detected voice sounds and processes only voices from a defined range and field of view relative to microphone array **310**. As described above, the identification of voice range and vector is based upon triangulation from differences in time of reception of the voice sounds at each microphone. The filtering of voices outside of the defined range and field of view is applied based upon the comparison of received voices at each microphone of microphone array **310**. Once the voice logic of processing resource **312** is complete, the voice sounds isolated in the desired distance and range are provided to a second processing resource **314**, such a second DSP, to remove nonhuman sounds. The processed speech is then communicated from the audio system, such as by wireless personal area network signals **316**, i.e., with a BLUETOOTH interface of keyboard **32** to an information handling system managing a videoconference. Although multiple DSP's are described in the above example embodiment, the voice location, selection and filtering may be performed by a single DSP.

(60) Referring now to FIG. **19**, a flow diagram depicts a process for managing voice sounds captured at a keyboard microphone array for use in an audio or video conference supported by an information handling system. The process starts at step **320** with reception by an array of three or more microphones of voice sounds conditioned on detection within a range of one meter or less and an angular field of view coverage of within 120 degrees of the front face of a keyboard. At step **322**, as a single speaker voice is detected, the three microphones of the array detect the voice with a different phase to determine a distance of the voice sound travel to each microphone, such as based upon a defined angular and distance separation between the microphones. At step **324**, a first artificial intelligence logic executing on a first DSP processing resource calculates the average time each microphone receives the sound based upon voice wave phase differences. At step **326**, based upon the three microphone average reception times, the artificial intelligence logic estimates the angle and distance of the speaker position relative to the microphone array. At step **328**, the artificial intelligence logic compares the converged location of the speaker voice to the predefined range of non-filtered voice, such as a range of less than one meter and within 120 degrees of the front of the keyboard. At step **330** a determination is made of whether an identified speaker voice falls within the defined range, such as one meter and 120 degrees of the front of the keyboard. If the voice is outside of the defined range, the process continues to step **332** to reject the voice. If at step **330** the voice is within the defined range and field of view, the process continues to step **334** to accept the voice and process the speech. At step **336** a determination is made of whether the accepted speech has sufficient quality. If not, the process continues to step **340** to provide a visual

indication to the end user of poor speech quality so that the end user can take appropriate steps to improve the quality. If at step **336** the quality is sufficient, the process continues to step **338** to indicate at the display a good speech level and to step **342** to output the audio to an information handling system, such as through a BLUETOOTH interface of the keyboard.

(61) Referring now to FIG. **20**, an example embodiment depicts presentation at a keyboard **32** integrated display **346** of audio capture conditions for a microphone array integrated in keyboard **32**. In the example embodiment, display **346** is an LCD display that depicts microphone and speaker controls when an information handling system application is using the keyboard to capture audio in support of a conference. At each microphone location a microphone usage state light **352** is illuminated to show when each microphone is active, such as with a white light when active and a red light when inactive. Microphone usage state lights **352** may be presented as part of the display presentation or may be a separate LED disposed under the display to illuminate when appropriate. A dynamic volume bar **348** presented at the upper portion of display **346** visually depicts the quality of the audio captured by the microphones. For instance, the DSP processing the audio or an MCU supporting the DSP and wireless communication interfaces with the display through a GPIO that outputs a speech quality indicator. In the example of FIG. **20**, three white lights indicate active operation of the microphones and a full dynamic volume bar indicates good quality audio is being captured. For instance, the speaker is within the 1 meter and 120 degree field of view zone and is speaking with adequate volume. Control icons **350** provide an end user with controls to manage a video conference, such as turning on and off a camera, muting audio capture, changing speaker volume, ending the call, etc.

(62) Referring now to FIG. **21**, keyboard **32** has a display **346** that presents low audio capture quality. In the example, microphone usage state lights **352** may turn a different color, such as red, or may flash to indicate microphone difficulties. Dynamic volume bar **348** shows a decreased length that corresponds to a level of quality captured by the microphones. A head icon **340** depicts an up arrow to suggest that the end user speak in a louder voice. In addition, information may be presented at the display showing the position of detected speakers relative to the zone in which voice is accepted and captured for use in a conference audio. For example display **346** may present an estimate location of detected speakers relative to a zone of captured speech to alert the end user that some speakers are being filtered out and to allow the end user to select a larger zone or to turn off the filtering so that all audio is included in the captured audio.

(63) Referring now to FIG. **22**, a flow diagram depicts a process for indicating at a keyboard display a quality of audio captured by the keyboard microphone array. The process starts at step **362** with a microphone signal **362** communicated to a DSP processing resource **360**. At step **364**, artificial intelligence executing on the DSP processing resource isolates desired speech audible sound from speech outside of the desired zone and from non-human sounds. At step **366**, level detector logic analyzes the captured sounds to determine the quality of the captured sounds for presentation at the keyboard display. At step **368**, a mapping of the captured speech quality is provided to applications executing on the information handling system that are using the speech, such as a mapping to ZOOM or TEAMS of loudness levels of the captured speech. In addition, the mapping can provide estimates of locations of speakers relative to the keyboard for presentation by the application and at the keyboard display, including a location of speakers who are filtered out of the captured audio. At step **370** a signal power calculation is performed on the captured audio to determine the speech volume. At step **372**, an envelope detection analysis is performed to determine the speaker locations. At step **374** a moving average control is applied to the volume and location analysis to smooth the calculated data. At step **376**, the volume and location of the captured audio is compared against a quality threshold. A GPIO of the DSP processing resource outputs the quality to an MCU processing resource **378** of the keyboard for analysis and presentation at the keyboard display. Based upon the DSP GPIO output, instructions executing on the MCU processing resource determine a light status for the microphones. At step **380**, if the

captured audio is above the audible threshold, the process illuminates a white color at step **382** for each of the microphone light indicators. If the captured audio at step **380** is below the audible threshold, the process illuminates a red color at step **384**.

(64) Referring now to FIG. **23**, a block diagram depicts a system for managing a keyboard, video, mouse (KVM) switch **402** of a display through a keyboard integrate display **346**. Display **30** includes a KVM switch **402** that interfaces with keyboard **32** and mouse **34** through cabled connections, such as USB cables, to coordinate use of the keyboard and mouse with each of plural information handling systems **10**. For instance, KVM switch **402** is managed by a scalar **400** of display **30** so that an end user can interact through an onscreen display menu (OSD) of display **30** to select which information handling system **10** presents visual images at display **30** and receives end user inputs from the keyboard and mouse. KVM switch **402** is a convenient tool for situations where an end user has multiple information handling systems in concurrent use, such as a work station having a desktop system where the end user also brings a portable system. The conventional end user interface with KVM switch **402** selects which information handling system **10** is active through buttons or a joystick of display **30** that controls an onscreen display menu presented by scalar **400**. This type of control is awkward for an end user who has to reach to the display and navigate the onscreen display menu manually over top of the visual images presented by the active information handling system. To simplify KVM switch interactions, display **346** of keyboard **32** presents a KVM user interface with a KVM manager **406** stored in non-transitory memory of MCU processing resource **404** that executes in cooperation with scalar **400**. Although the example embodiment depicts a wired connection between the keyboard, mouse and the KVM switch, alternative embodiments may use a wireless interface through WNIC **24**.

(65) To present KVM switch control at keyboard display **346**, KVM manager **406** duplicates the display onscreen menu of scalar **400** with its KVM functionality for touch interactions through keyboard display **346**. The end user is provided an icon on display **346** to activate the KVM menu at display **346** so that control of selection of an active information handling system is at the end user's fingertips with touch inputs to the onscreen display menu through keyboard display **346**. This allows an end user to readily toggle between information handling systems without disruption of the display presentation by the onscreen display menu at display **30**. KVM manager **406** may present the KVM control menu in a variety of ways. In one example embodiment, a copy of the KVM control menu is stored in flash memory of MCU processing resource **404** and activated by a touch at an icon that responds to activation by showing information handling systems available to select with an additional touch. Selection of an information handling system for interaction with the KVM switch at display **346** is communicated through the keyboard wireless or wired interface and performed by scalar **400**, such as through an I2C or other interface.

(66) Referring now to FIGS. **24**, **25** and **26**, example embodiments depict the type of KVM control available through a keyboard touchscreen display. FIG. **24** depicts first and second information handling systems **10** that each interface with display through a video cable **28** and a USB cable **390** or similar data interface. An end user toggles an icon presented at keyboard **32** to alternatively interface keyboard **32**, mouse **34** and display **30** with each information handling system **10**. FIG. **25** depicts an alternative presentation of visual images at display **30** that an end user can select from the keyboard display. In the example embodiment, display **30** shows a first window **392** having visual information from a first of the two information handling systems **10**, and a second window **394** having visual information from a second of the two information handling systems **10**. In this example embodiment, display **30** may alternate control of keyboard **32** and mouse **34** based upon which of windows **392** and **394** are active, or the active information handling system may be selected through the keyboard display. FIG. **26** depicts another alternative presentation having keyboard **32** and mouse **34** interfaced through one display **30** and a second display **30** having a daisy chain interface with the first display. In this example embodiment, communications between the displays **30**, such as with the DisplayPort protocol, coordinates selection of KVM switch

interfaces with each of the information handling systems **10**. For instance, the displays may both present visual images from one information handling system **10** that also interfaces with the mouse and keyboard. Alternatively, each display may present visual information of one of the information handling systems **10** while keyboard and mouse control is managed by selections made at a display of keyboard **32**. In various embodiments, control of KVM switch interfaces may be managed with instructions stored and executed on keyboard **32** that command KVM actions, by instructions of a scalar of a display **30** that defines the interface shown at the keyboard display, and/or instructions of an operating system on one of the information handling systems that provides the keyboard display control user interface based upon the type of displays found by the operating system.

(67) Referring now to FIGS. **27A** and **27B**, a flow diagram depicts an example embodiment of keyboard touchscreen KVM control and associated use interfaces. The process starts at step **410** with one keyboard connected directly to a first monitor through a USB cable, USB hub, wireless dongle or other interface. At step **412** an end user controls a first information handling system and first display having an integrated KVM switch with the keyboard through a USB upstream interface from the display to the information handling system. For instance, keyboard inputs are communicate to the display KVM switch and from the KVM switch to the first information handling system. At step **414**, the end user decides to change the keyboard to interface with a second information handling system with a command provided through a touchscreen display integrated in the keyboard. In the example embodiment, a user interface **416** illustrates an example of a keyboard display user interface having an “OSD” icon that the end user presses to initiate presentation of an onscreen display menu with controls to select KVM switch interfaces to the first and second information handling systems. At step **418**, in response to selection of the OSD icon, the keyboard display presents an OSD menu having a graphical user interface with commands available for selection of KVM switch interfaces. For instance, the graphical user interface **420** presents OSD menu options stored in non-transient memory of the keyboard. In the example embodiment, the end user can manage all of the display OSD menu items, such as brightness and contrast. At step **422**, the end user selects the KVM switch icon to initiate control of the KVM switch. In response at step **422** of a press at the OSD menu icon, the keyboard display graphical user interface **424** presents graphical user interface **426** providing the end user with a selection of the first or second information handling system to receive the KVM switch upstream communications from the keyboard and mouse.

(68) Upon toggling from the first information handling system to the second information handling system at step **422**, the process continues to step **428** at which the first display receives the upstream toggle OSD command from the keyboard display. At step **430**, the USB hub of the KVM switch converts the onscreen display command to an I2C signal for application at the KVM switch. The process continues to step **432** at which the first display scalar receives the I2C command and routes the command to the onboard OSD processing resource, such as an integrated circuit microcontroller unit included in the first display. At step **436** the first display KVM processing resource receives the command to switch the upstream destination for the keyboard and mouse inputs directed to the KVM switch and, at step **438** commands a change in the destination of the mouse and keyboard inputs to the second information handling system. Once the keyboard display graphical user interface changes the information handling system that is active, the process continues to step **440** at which the end user can select a graphical user interface **442** that presents a quick toggle capability to command a change in the KVM switch configuration with a single touch. The quick toggle icon of graphical user interface **442** may remain active and presented at the keyboard display automatically when the keyboard detects an interface to an information handling system through a KVM switch.

(69) Referring now to FIG. **28**, an example embodiment depicts management of dual display and dual information handling system KVM interfaces with a keyboard display. At step **444**, the keyboard **32** connects via a USB interface, such as through a wireless dongle **450**, to a first display

30 and first information handling system **10**. In the example embodiment, keyboard **32** inputs are communicated through the KVM switch of the first display **30** through and upstream **1** interface and visual information is provided from the first information handling system **10** through a video **1** cable for presentation at the first display **30**. The KVM switch in the first display coordinates interactions with the first information handling system so that at step **444** keyboard **32** inputs are sent to the first information handling system at step **446**. In addition, the second information handling system **10** interfaces with a second display **30** through a second video cable and with the first display through an upstream cable that is not active since the first display KVM is communicating keyboard inputs to the first information handling system. Keyboard **32** includes integrated touchscreen display **346** that presents a graphical user interface to control selection of KVM switch communication configuration for the KVM switch included in the first display. At step **448**, an input at a graphical user interface presented by keyboard display **346** commands a change in control at the KVM from the first information handling system to the second information handling system with an input at the KVM icon of the onscreen display menu options.

(70) In response to the command from the keyboard display **346**, at step **452** the KVM switch reconfigures so that keyboard inputs communicate from KVM switch of the first display to the second information handling system. In the example embodiment, each information handling system **10** continues to present visual images at the display to which the video of the information handling system is coupled, however, keyboard **32** key inputs are directed to each information handling system individually so that, at step **452**, keyboard inputs communicated to the second information handling system will be shown at the second display and will be used for execution on the second information handling system. At step **452**, keyboard inputs are communicated to the second information handling system **10** for use by the second information handling system at the second display at step **454**. In an alternative embodiment, the upstream cable connection between the first monitor and the second information handling system may include a daisy chain or other downstream video capability so that keyboard display **346** can also command a change at the KVM of the first display regarding presentation of visual images by the first or second information handling system.

(71) Although the present invention has been described in detail, it should be understood that various changes, substitutions and alterations can be made hereto without departing from the spirit and scope of the invention as defined by the appended claims.

Claims

1. An information handling system comprising: a housing; a processor coupled in the housing and operable to execute instructions to process information; a memory coupled in the housing and interfaced with the processor and operable to store the information and instructions; a display interfaced with the processor and operable to present the information as visual images; and a peripheral keyboard separate from the housing and interfaced with the processor, the keyboard having keys to accept keyed inputs, an array of at least three microphones arranged in a triangle configuration and operable to capture sounds as audio information and a processing resource included in the keyboard and interfaced with the array of microphones, the processing resource analyzing the audio information by triangulation to filter out voices in the audio information that originate outside of a predetermined range and field of view relative to the keyboard.
2. The information handling system of claim 1 wherein the range comprises one meter and the field of view comprises within 120 degrees of the front of the keyboard.
3. The information handling system of claim 1 wherein the keyboard processing resource comprises a digital signal processor having a first filter that removes voice sound information from outside of the predetermined range and field of view and a second filter that removes all non-voice sound information.

4. The information handling system of claim 1 further comprising: a keyboard display included in the keyboard and interfaced with the processing resource, the keyboard display operable to present information as visual images; and a user interface presented at the keyboard display visually depicting a quality of the audio information captured by the array of microphones.
5. The information handling system of claim 4 wherein: the user interface depicts one or more adjustment options to the filter predetermined range and field of view; and the processing resource applies an input of the one or more adjustment options to adjust the range and field of view of the audio information filter.
6. The information handling system of claim 5 wherein: the user interface depicts a relative location of a voice sound audio information origination location, the voice sound audio information filtered by the processing resource as outside the predetermined range and field of view; and the user interface depicts an option to unfilter the voice sound audio information.
7. The information handling system of claim 4 wherein the keyboard display presents a first color at a location of the microphones of the microphone arrays when audio information sound quality exceeds a threshold and a second color when the audio information sound quality fails to exceed the threshold.
8. The information handling system of claim 4 wherein the user interface depicts suggestions to improve sound quality when the sound quality is below a threshold.
9. The information handling system of claim 1 wherein microphone array comprises three microphones having a defined spacing and the processing resource detects sound wave phase difference at the three microphones to locate voice origination.
10. A method for managing audio information of an information handling system, the method comprising: capturing audible sounds as audible information with an array of at least three microphones included in a keyboard and arranged as a triangle; analyzing the audible information with a processing resource included in the keyboard using triangulation to identify voice of the audible information that originates at a location within a predetermined zone relative to the keyboard and voice of the audible information that originates outside the predetermined zone; and filtering the voice that originates outside of the predetermined zone from the audible information.
11. The method of claim 10 further comprising: filtering all non-voice sounds from the audible information.
12. The method of claim 10 wherein: the array of microphones comprise three microphones coupled to the keyboard with a predetermined spacing; and the analyzing comprises comparing the phase of audible sounds captured by each of the three microphones.
13. The method of claim 10 further comprising: coupling a display to the keyboard; and presenting at the keyboard display quality information related to the audible information.
14. The method of claim 13 further comprising: presenting at the keyboard display a user interface to control capture of audible information by the array of microphones; and adjusting the size of the predetermined zone to adjust the voices filtered from the audible information.
15. The method of claim 13 further comprising: presenting at the keyboard display an indicator of a first color when the audio quality exceeds a threshold; and presenting at the keyboard display an indicator of a second color when the audio quality is less than the threshold.
16. The method of claim 13 wherein the quality information comprises a bar presented at the display of variable length based on audible information quality.
17. A peripheral keyboard comprising: a peripheral keyboard housing; plural keys coupled in the peripheral keyboard housing to accept keyed inputs; an array of at least three microphones coupled in the peripheral keyboard housing and operable to capture sounds as audio information; a processing resource coupled in the peripheral keyboard housing and interfaced with the array of microphones, the processing resource analyzing the audio information by triangulation to filter out voices in the audio information that originate outside of a predetermined range and field of view relative to the keyboard.

18. The peripheral keyboard of claim 17 further comprising: a keyboard display interfaced with the processing resource, the keyboard display operable to present information as visual images and having a touchscreen to accept end user inputs; and a user interface presented at the keyboard display visually depicting a quality of the audio information captured by the array of microphones.
19. The peripheral keyboard of claim 18 wherein the processing resource comprises a digital signal processor having a first filter that removes voice sound information from outside of the predetermined range and field of view and a second filter that removes all non-voice sound information.
20. The peripheral keyboard of claim 19 wherein: the user interface depicts one or more adjustment options to the filter predetermined range and field of view; and the processing resource applies an input of the one or more adjustment options to adjust the range and field of view of the audio information filter.
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